

An Evaluation of SociusBot

A chatbot-based roleplay partner and tutor that assists students in training their social skills primarily based on pragmatical ability.

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Thank you for participating in our study!

Please read through the overview of our study and the procedure below before starting. Our research assistants will guide you through the evaluation process. If you have any questions about this evaluation process, please send us an email using the addresses listed above.

Overview

SociusBot is an AI-powered chatbot designed to support young adults in developing pragmatic communication skills through guided conversational practice. The chatbot engages users in realistic social scenarios grounded in Filipino cultural values and the developmental challenges of emerging adulthood, helping them navigate difficult conversations with clarity, respect, and cultural sensitivity.

SociusBot is intended for Filipino young adults aged 15–24 years old who are currently enrolled in senior high school or undergraduate programs. For this study, we would require at most 20 participants who are comfortable communicating in English and navigating digital applications.

Conversation logs will be collected to monitor how users engage with SociusBot. These logs will include users' responses during scenario-based conversations and their personalized tutoring conversations. These will be used strictly for research report generation.

Procedure

To participate in this study, we will require **one (1) week** of your time for you to go through the following steps:

- **Orientation (40 minutes).** This gives you an overview of the study, the informed consent and informed assent for minor participants, the experiment proper, and the debriefing. You can raise questions to be addressed during this orientation. The orientation may be done in groups or individually depending on your availability and preference. The orientation will be conducted online.
- **Informed Consent (30-60 minutes).** You will be given time to review the Informed Consent which outlines voluntary participation, withdrawal, data privacy and confidentiality, and how your data will be used in the study.
- **Pre-Assessment Survey (15 minutes).** Before beginning conversations with the chatbot, you will complete an initial survey to assess your baseline perceptions and communication skills.

- **Conversations with the Chatbot (1 week, minimum 1 conversation per day).** You will engage in conversations with the chatbot for a period of one (1) week. You are required to complete at least one (1) conversation per day, choosing from one of the four scenario themes provided. Each conversation should last at least 10-15 minutes. All conversations will be recorded for later analysis, but no video recording or participant observation will be conducted throughout this period.
 - **Weekly Reflection Survey (15 minutes per week).** At the end of each week, you will complete a brief online survey to reflect on your experience using the chatbot that week, including your perceived development in pragmatic ability, feedback comprehension, and scenario authenticity.
 - **Final Reflection Survey and Chatbot Evaluation (30 minutes).** At the end of the three-week period, you will complete a comprehensive survey evaluating your overall experience, development, and impression of the chatbot.
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Research Instruments

Five (5) research instruments will be used during this activity – the *Informed Consent Form*, the *Informed Assent Form*, the *Pre-Assessment Survey*, the *Daily Post-Interaction Survey*, and the *Consolidated Evaluation Survey*. They are attached to this document.

Informed Assent Form

By filling out this Informed Consent Form, I signify that:

- ☐ I have read the overview of the research and the procedure for the evaluation process.
- ☐ I agree to allow my child/student to participate in this study on developing pragmatic competence using SociusBot.
- ☐ I understand that my child/student will interact with SociusBot through scenario-based conversations, complete weekly reflection surveys, and provide feedback through the pre-assessment, weekly, and final reflection surveys.
- ☐ I have reviewed the study procedure, including the 1-week experimentation period requiring a minimum of one (1) conversation per day.
- ☐ I understand that my child/student is under no obligation to take part in this research, and have the right to withdraw at any time without consequences.
- ☐ I understand that data collected from my child/student will be anonymized and aggregated to produce the research report.
- ☐ I understand that findings from this research may be published in a research journal or conference paper, and disseminated in a conference.
- ☐ I agree to have the conversation logs containing the conversations held by my child/student with SociusBot be used for education and research purposes on the condition that our privacy is respected.
- ☐ I understand that neither my name nor my child/student's name will appear in any publication.
- ☐ I understand that all conversation logs will be anonymized, with names of participants, families, peers, and other persons replaced with placeholders.
- ☐ I understand that if my child/student exhibits signs of mental health-related symptoms, they will be advised to seek professional help and I will be informed accordingly.
- ☐ I understand that all data will be destroyed immediately after the end of the research.

Parent/Guardian : _____

Signature : _____

Date : _____

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Informed Consent Form

By filling out this Informed Assent Form, I signify that:

- ☐ I agree to participate in the SociusBot study and engage in conversations with the chatbot.
- ☐ I have been shown how to use SociusBot and understand that it will help me practice communication skills in realistic social scenarios.
- ☐ I understand that I will be asked to complete at least 3 conversations per week for 3 weeks, choosing from four different scenario themes.
- ☐ I understand that I will complete surveys before starting, at the end of each week, and at the end of the study to share my thoughts and experiences.
- ☐ I understand that I can stop participating at any time without any penalty.
- ☐ I understand that if I have questions or concerns at any time, I can ask the researchers.
- ☐ I understand that my conversations will be recorded but my identity will be kept anonymous and confidential.
- ☐ I understand that if I experience any discomfort or mental health concerns, I will be referred to professional help.
- ☐ I understand that giving my information means I agree to be in the study.

Name of Child / Student : _____

Age & Grade Level : _____

Pre-Assessment Form

Before we begin with our experiment, we would first like to know a little more about you.

Section 1

1. What is your current age?

- ☐ Below 15 years old
- ☐ 15-17 years old
- ☐ 18-24 years old
- ☐ Above 24 years old

2. Are you currently enrolled in any Philippine academic institution?

- ☐ Yes, in senior high school
- ☐ Yes, in an undergraduate program
- ☐ No

3. If yes, please provide the name of your institution: _____

4. If yes, what is your current program/year level?: _____

5. How would you rate your English proficiency?

- ☐ Beginner (I struggle with basic conversation)
- ☐ Intermediate (I can communicate but with some difficulty)
- ☐ Advanced (I can communicate comfortably in most situations)
- ☐ Fluent (English is one of my primary languages)

6. Are you comfortable using text-based chat applications (like Messenger, WhatsApp, etc.)?

- ☐ Yes
- ☐ No
- ☐ Somewhat

Section 2.

7. How comfortable do you feel in the following situations?

(1 = Very Uncomfortable, 2=Uncomfortable, 3=Neutral, 4=Comfortable, 5 = Very Comfortable)

	1	2	3	4	5
Speaking with authority figures (teachers, parents, supervisors).					
Making decisions independently.					
Having difficult conversations with friends.					
Expressing disagreement with a group.					

8. How often do you experience the following in conversations?

	1	2	3	4	5
Unsure how much information to share					
Difficulty understanding hints or indirect messages					

Staying on topic during conversations					
Being clear when explaining something.					

Have you experienced any of these challenges in social situations? (Select all that apply)

- ☐ Balancing personal goals with family expectations.
- ☐ Being treated as less independent than you feel
- ☐ Uncertainty in romantic or friendship relationships
- ☐ Conflicts between personal values and peer pressure
- ☐ None of the above

How interested would you be in using a chatbot to practice communication skills?

- ☐ Very interested
- ☐ Somewhat interested
- ☐ Neutral
- ☐ Not very interested
- ☐ Not interested at all

What concerns, if any, do you have about using a chatbot for communication practice? (Select all that apply)

- ☐ Privacy and data security
- ☐ Effectiveness of AI feedback
- ☐ Time commitment
- ☐ Chatbot feels unnatural or robotic
- ☐ Other: _____

What features would you find most valuable in a communication practice chatbot? (Rank 1-4, with 1 being most valuable)

- ____ Realistic conversation scenarios
- ____ Personalized feedback on my responses
- ____ Progress tracking over time
- ____ Ability to practice at my own pace

Daily Post Interaction Survey

Thank you for helping us test SociusBot! Your feedback is vital in improving SociusBot's affect and development to making social skills development more accessible. Please answer these questions honestly.

Instructions:

- Use the scale below to rate your experience:
😊 - Yes! I really liked it! 😊 - It was okay. 😐 - I'm not sure. 😞 - I didn't like it.
😡 - No! I didn't like it at all.
- For each statement, mark (✓) the box that best matches how you felt according to the scale above.
- At the end, you can write or tell us more about what you liked or didn't like.

Performance: How well did Socius, as a Conversation Partner, immerse you in a conversation?

Statement (Performance)	😡	😞	😐	😊	😄
1. Socius's story scenarios were easy to understand.					
2. Socius consistently stayed on topic.					
3. Socius steered me back on track if I went off topic.					
4. Socius adjusted in roleplay based on my answers					
5. Socius stayed in character so that I could fully immerse myself in the conversation					
6. It was easy to explore and move around Socius.					
Do you have anything else to say about how Socius immersed you in a conversation?					

Performance: How well did Socius, as an AI Mentor, provide you with relevant feedback regarding your conversations?

Statement (Performance)	😡	😞	😐	😊	😄
7. Socius's advice was easy to understand.					
8. It was easy to interact with the mentor.					
9. The feedback helped me correct my mistakes.					

10.Socius helped me understand how to build the right response based on the other person's response					
11.Socius adjusted the word problems to my skill level.					
12.Socius explained certain points in how I conversed and how to improve on them.					
13.It was easy to explore and move around Socius.					
Do you have anything else to say about how Socius provided you with feedback regarding your conversations?					

Humanity: Does conversing with Socius, as a Conversation Partner, feel natural? Does it converse like a real person?

Statement (Humanity)					
1. Socius's responses felt like a real human's.					
2. I could easily talk to Socius.					
3. The chatbot is very encouraging.					
4. The chatbot was patient and didn't rush me.					
5. Socius replied to my responses as expected of the role it is playing.					
Do you have anything else to say about Socius as a conversation partner? Did you feel like you were talking to a real person?					

Humanity: Does conversing with Socius, as an AI Mentor, feel natural? Does it converse like a real person?

Statement (Humanity)					
6. Socius's responses felt like a real human's.					
7. I could easily talk to Socius.					

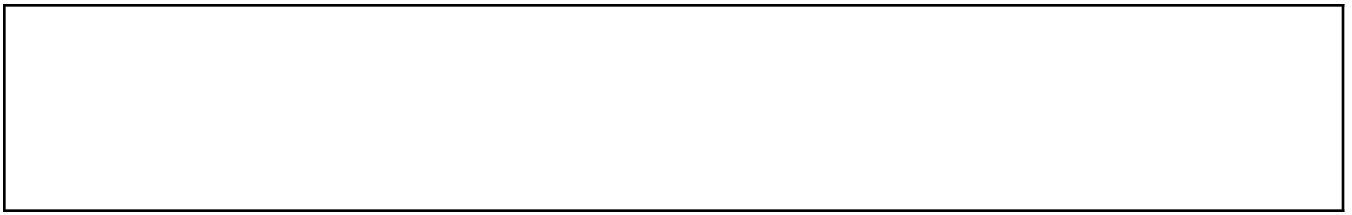
8. The chatbot is very encouraging.					
9. The chatbot was patient and didn't rush me.					
10. Socius answered all my questions properly.					
Do you have anything else to say about Socius as a mentor? Did you feel it was like a real mentor?					

Affect: Was Socius, as a Conversation Partner, fun and engaging?

Statement (Affect)					
1. I enjoyed conversing with Socius.					
2. Socius was fun to use.					
3. The chatbot made conversations in social scenarios less scary.					
4. I'd like to use Socius again.					
Do you have anything else to say about how Socius, as a conversation partner, made you feel?					

Affect: Was Socius, as an AI Mentor, fun and engaging?

Statement (Affect)					
1. I enjoyed receiving advice from Socius.					
2. Socius was fun to use.					
3. The chatbot made receiving advice less scary.					
4. I believe the chatbot's advice could help me in real world situations.					
5. I'd like to use Socius again					
Do you have anything else to say about how Socius as a mentor made you feel?					



SociusBot Consolidated Evaluation Form

This survey combines your weekly feedback and final evaluation. Please complete this survey at the end of the week (the end of the testing phase). Your honest feedback will help us improve Socius.

Scale for all Likert items:

1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

Section 1: Overall Experience

- 1.1 My overall experience with SociusBot throughout the program was positive.
- 1.2 The chatbot consistently felt like a real person over the course of the testing phase.
- 1.3 The conversation scenarios remained engaging and interesting every day.
- Open-ended: Describe what stood out to you most about your overall experience using SociusBot.

Section 2: Learning & Behavioral Change

- 2.1 I feel more confident in my social communication skills now than before I started.
- 2.2 Socius helped me understand how to build better responses based on others' cues.
- 2.3 I believe using Socius regularly can lead to long-term social improvement.
- Open-ended: Describe a moment where you felt challenged during a conversation and how you handled it. Have you successfully applied any of SociusBot's strategies in real-life conversations?

Section 3: AI Mentor & Feedback Quality

- 3.1 The AI Mentor's feedback was consistently helpful across all sessions.
- 3.2 The explanations of Grice's Maxims (conversation rules) and implicatures were clear and easy to grasp.
- 3.3 The feedback I received felt personalized to my specific communication style.
- Open-ended: What specifically made the AI Mentor's feedback helpful or unhelpful for your learning?

Section 4: Scenario Design (Pragmatic Competence)

- 4.1 The scenarios felt grounded in realistic cultural and developmental frameworks.

- 4.2 The themes (e.g., Identity vs. Expectation, Autonomy vs. Dependence) felt relevant to my life as a young adult.
- Open-ended: Which scenario theme felt most relevant to you, and are there any specific situations you would like to see added?

Section 5: Emotional Safety & Usability

- 5.1 I felt safe to make mistakes and experiment without fear of judgment.
- 5.2 I never felt discouraged or criticized by the AI Mentor's feedback.
- 5.3 The interface was consistently easy to navigate and I rarely felt confused by how it worked.
- 5.4 The system was technically reliable and I did not encounter bugs or glitches.

Section 6: Final Reflections

- 6.1 I would recommend Socius to my peers or friends.
- 6.2 I would be interested in using Socius again in the future.
- Open-ended: What was the single most important thing you learned or liked about Socius?
- Open-ended: If you could change one thing to make Socius more helpful or less frustrating, what would it be?